SMART INDIA HACKATHON • IDEA PRESENTATION

ORCA

Agentic AI for Ocean Intelligence

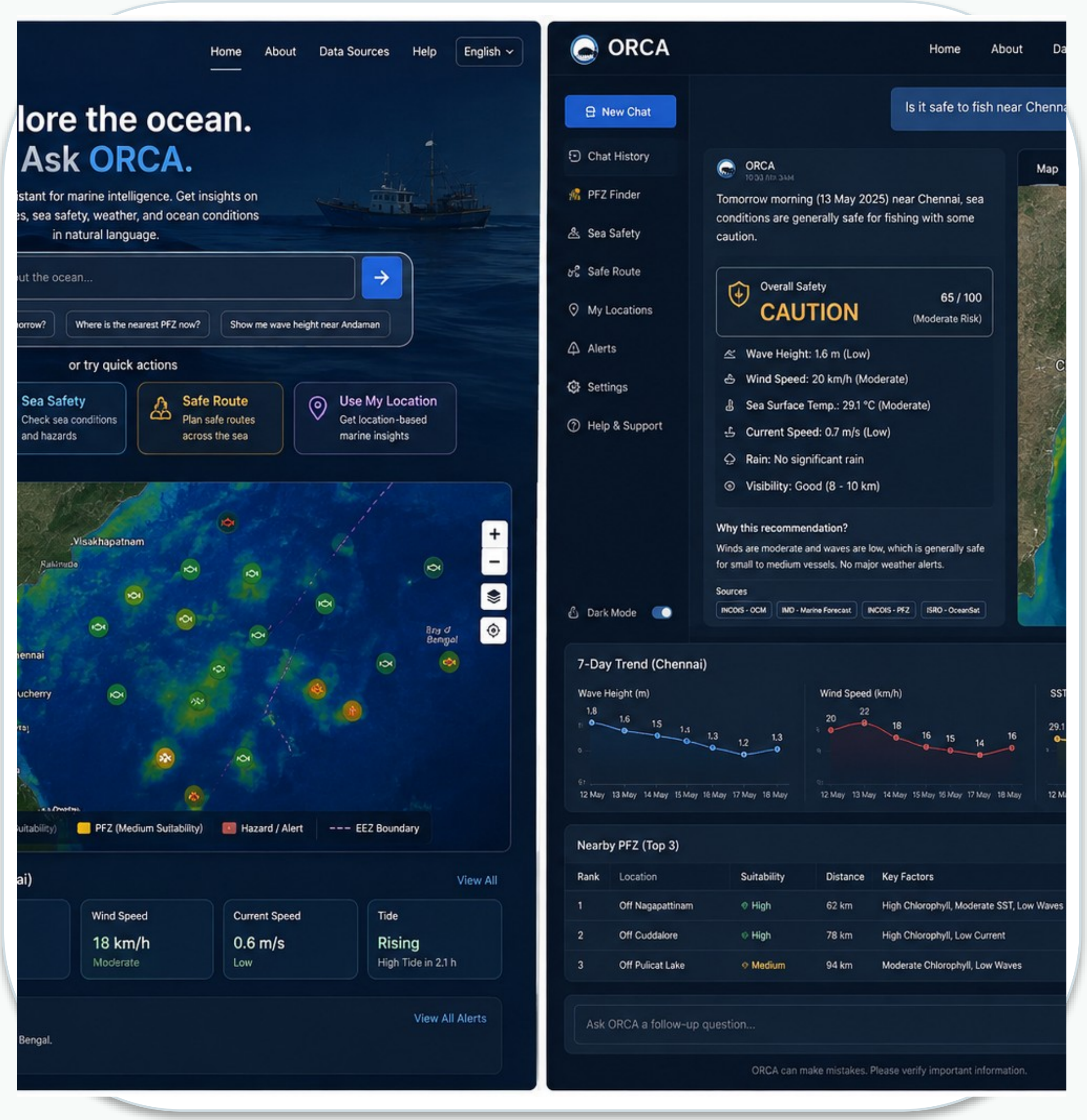
Ask about the ocean in simple language. ORCA connects the right information, analysis and maps to help you decide what to do next.

SIH26176 • SOFTWARE • MISCELLANEOUS

Problem: ORCA Marine EcOsystem Reasoning with Collaborative Agents

Team ID: ___________ Team Name: CodeCatalysts

Marine • Weather • Satellite • Ocean • GIS • AI



Proposed Solution

Keep the three required SIH sections. Use evidence and user groups to make the problem real.

1. DETAILED SOLUTION

ORCA is a single marine intelligence assistant.

Users ask questions by text or voice. ORCA understands the need, brings in the relevant information, and returns a clear answer with a map, alerts, routes or evidence.

2. HOW IT ADDRESSES THE PROBLEM

Instead of asking users to search many platforms, ORCA connects the right sources and analysis for the question.

The result changes with the user, location and time.

3. INNOVATION & UNIQUENESS

Eight specialised agents work together rather than one general chatbot.

ORCA combines AI reasoning with geospatial logic and shows supporting evidence when available.

11,099 km	18%	~80M	28M+	95%
coastline	population in 72 coastal districts	people depend on the sea	fishermen in India	trade by volume via maritime routes

WHO NEEDS IT?

- Fishing communities
- Researchers
- Authorities
- Maritime operators
- Environment

SLIDE 3

Technical Approach

A simple view of how the system moves from a user's question to a decision.

IMPLEMENTATION FLOW

1. **USER**
   Question + optional location
2. **PLANNER**
   Understands request + selects agents
3. **8 AGENTS**
   Work together on the task
4. **DATA**
   Marine + weather + satellite + GIS
5. **VALIDATE**
   Check source, time and location
6. **REASON**
   Assess conditions + risk
7. **OUTPUT**
   Chat + map + alerts + route + evidence

ORCA SYSTEM ARCHITECTURE

TECHNOLOGY STACK

Frontend
HTML • CSS • JavaScript
Leaflet maps

Backend
Python • FastAPI
Uvicorn

AI
Anthropic Claude
8-agent orchestration

Database / GIS
Supabase
PostgreSQL + PostGIS

Feasibility, Validation & Scalability

We can build it in steps, test it with real scenarios, and expand it over time.

FEASIBILITY

Data — Multiple marine, weather, satellite, ocean and GIS sources.

Technology — Existing AI, web and geospatial tools.

Modular design — Agents and data connectors can be added independently.

VALIDATION

1. Prototype
2. Real data
3. Test scenarios
4. Compare
5. Measure
6. Feedback

WHAT WE WILL CHECK

- Response time
- Data freshness
- Source consistency
- Output relevance

SCALABILITY

MVP — Core features + key data

Pilot — One coastal region

Regional — More states + datasets

National — Wider marine platform

RISKS → FIX

Different formats → common data layer • Data delay → timestamp + freshness • Conflicts → cross-check • LLM errors → grounded data

Feasible today • Validated with real scenarios • Scalable as data, agents and users grow

SLIDE 5

Impact & Benefits

ORCA is useful to people who work with, study, manage or depend on the ocean.

1

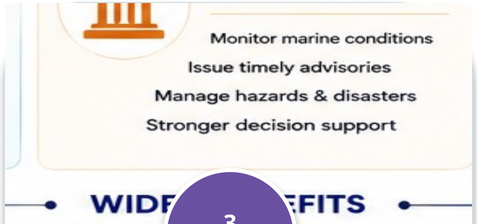
FISHIN UNITIES

- Find better fishing zones
- Plan safer trips
- Receive timely alerts

2

RESEARC ACADEMIA

- Access multi-source data
- Study trends & patterns
- Support ocean research



3

GOVERNM UTHORITIES

- Monitor hazards
- Support coastal response
- Improve decisions

4

MARIT RATORS

- Plan safer routes
- Check sea conditions
- Reduce operational risk

5

ENVIRONM AKEHOLDERS

- Monitor sensitive areas
- Support conservation
- Promote sustainable use

SOCIAL	ECONOMIC	ENVIRONMENTAL
Safer, easier access to marine information	Better planning, lower avoidable effort	Better awareness of sensitive areas

~80M	28M+	95%	₹62,408 Cr
sea-dependent livelihoods	fishermen	trade by volume	seafood exports

Scale is shown through national relevance today and the ability to grow from pilot regions to wider marine operations.

Research & References

Keep this slide clean: what we study, what kinds of data we use, and where the evidence comes from.

RESEARCH AREAS

Marine Earth Observation
Satellite-based ocean observations

Ocean & Weather Forecasting
Waves, wind, temperature, sea conditions

Geospatial Analysis
GIS, location and spatial reasoning

Multi-Agent AI
Planning, collaboration and reasoning

Decision Support
Risk, alerts, recommendations and maps

DATA SOURCE CATEGORIES

Satellite & Ocean
SST • chlorophyll • ocean observations

Weather & Forecasts
Wind • waves • marine conditions

Buoys & Sensors
In-situ observations

Marine Advisories
Warnings • fisheries • hazards

GIS & Boundaries
EEZ • protected/restricted areas

Historical Data
Past observations and time series

KEY REFERENCES

INCOIS
Indian ocean observations, forecasts and advisories

NOAA / CoastWatch
Satellite ocean data and imagery

NOAA / OceanWatch
Ocean and environmental observations

WaveWatch III
Global wave model products

Leaflet
Interactive web maps

Supabase + PostGIS
Database and spatial data

Final submission: replace these source names with the exact verified dataset pages, APIs and research papers used in the implementation.